SUPPORTING ADULT GUIDANCE

Bounds in problems

Percentages, Ratio, Growth and Bounds · Lesson 10. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The lesson opens with a bridge (4 m to the nearest metre) and a lorry (3.8 m to 1 d.p.): can we be sure it fits? Movement is a quick yes, no or can't-be-sure round. Reflection has six questions and a stretch on why an answer can only be as accurate as its bounds agree. The main activity, the How Sure? Workshop, is in Creativity. Learners choose a problem (a speed, a density and whether it floats, flooring for a room, tins of paint), set each measurement and its accuracy, and see the exact bounds of the answer. A ladder rounds both bounds to 1, 2, 3 and 4 significant figures and marks where they agree; learners choose the accuracy to give the answer to and get feedback. A question at the foot (under the limit? enough? does it float?) gives yes, no or can't be sure, with the interval drawn against the limit. Six briefs follow, including making a decision certain by measuring more precisely.

Driving Question: *How sure can I be of this answer?*

The mathematics, in plain terms

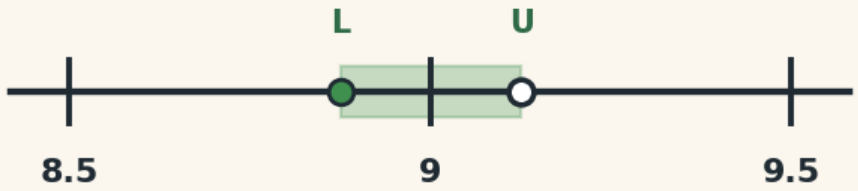
An answer is certain to k significant figures exactly when every value in $[L, U)$ rounds to the same value. The upper bound is not reached, so the ladder rounds the values just below U (on an exact halfway point they round down). Agreement is usually lost as k increases, but not always: an interval such as 12.497 to 12.503 straddles 12.5 (so it cannot be given to 2 s.f.) yet lies inside 12.45 to 12.55 (so it can be given as 12.5 to 3 s.f.). The workshop reports the most accurate degree that is certain. Examination questions use the simpler rule (round both bounds and compare); the two agree except at such boundaries.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — A suitable degree of accuracy

A cyclist covers 360 m in 40 s, both to the nearest whole number. Give the speed to a suitable degree of accuracy. (On the line, *L* and *U* mark the lower and upper bounds of the speed.)



1. Lower bound: smallest over largest $\frac{359.5}{40.5} = \frac{719}{81} = 8.88$ to 2 d.p.

2. Upper bound: largest over smallest $\frac{360.5}{39.5} = \frac{721}{79} = 9.13$ to 2 d.p.

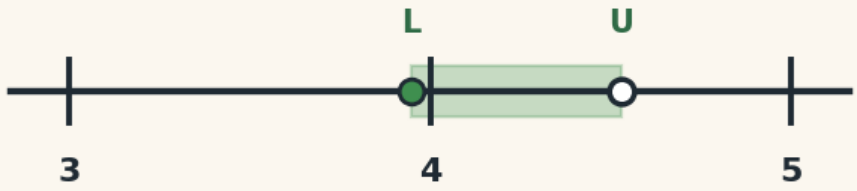
3. To 2 s.f. 8.9 and 9.1: different

4. To 1 s.f. 9 and 9: the same

9 m/s, to 1 significant figure. (Both bounds lie between 8.5 and 9.5, so every possible speed rounds to 9.)

Example 2 — Is it enough?

A wall is 38 m^2 and one tin of paint covers 9 m^2, both to the nearest m^2. How many tins are sure to be enough? (*L* and *U* mark the fewest and the most tins you could need.)



1. The most tins you could need	$\frac{38.5}{8.5} = 4.53$ to 2 d.p.
2. The fewest you could need	$\frac{37.5}{9.5} = 3.95$ to 2 d.p.
3. Decide	4 tins might not be enough; 5 always are

5 tins are sure to be enough.

Decide, then say how accurately

Learners first decide yes, no or can't be sure from intervals they can picture, then build the bounds of real answers and read how far the bounds agree. The briefs turn this round: measure more precisely until a decision becomes certain.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Will the lorry fit? The idea card.	~4 min
Movement	Can you be sure? Eight quick decisions.	~5 min
Reflection	Six questions; the stretch on accuracy.	~10 min
Creativity	The How Sure? Workshop (main activity) and six briefs.	~21 min
Rest	“I can't be sure yet” is an honest answer.	~2 min
Nutrition	Practical: knowing how much to trust a number.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Giving the answer from the rounded values to many figures.
Ask: “Could it really have been that precise?”

Deciding from one bound only.
Ask: “What about the other end of the interval?”

Planning for the lower bound when asking “is it enough?”
Ask: “What is the most you could need?”

Rounding the bounds of the measurements before calculating.
Ask: “Can you keep the exact bounds until the end?”

Writing the answer to the accuracy where the bounds first differ.
Ask: “Do both bounds round to the same value at this accuracy?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: calculating with bounds (Lesson 9) and rounding to significant figures. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner gives a speed found from 360 m and 40 s (both to the nearest whole number) as 9.0 m/s. The aim is a question about how sure the 0 is.

- “What are the bounds of the speed?”
- “What do they round to, to 2 significant figures?”

- “Are they the same?”
- “So how many figures can you be sure of?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why does the workshop say “values near the upper bound”?
The upper bound is never reached. If it sits exactly on a rounding boundary, the values just below it round down, so the workshop rounds those values.

Can an answer be certain to 3 s.f. but not to 2 s.f.?
Occasionally, yes, when the interval straddles a 2 s.f. boundary but lies inside one 3 s.f. interval. The workshop reports the most accurate certain degree and explains the gap.

Is the stretch required?
No. The secure core is finding both bounds of the answer and using them to decide and to choose the accuracy.

What comes next?
This is the last lesson of the unit. Sets and Venn Diagrams follow.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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